

Normal-Fixed Point (NFP) Equation

Problem Formulation:

Given: $K \subseteq \mathbb{R}^n$ closed convex, $F: K \rightarrow \mathbb{R}^n$

Find: $\bar{x} \in K$ such that $(x - \bar{x})^T F(\bar{x}) \geq 0$ for all $x \in K$

Reformulation as Fixed Point:

$$\begin{aligned}\Psi_\alpha(x) &= f(P_K((1 + \alpha)x - \alpha P_K(x - \alpha f(x)))) \\ &+ \alpha(P_K(x - \alpha f(x)) - P_K(x)) = 0\end{aligned}$$

Properties of Ψ_α :

- Combines projection and fixed point formulations
- Allows use of fixed point iteration theory
- Parameter $\alpha > 0$ controls stepsize

Key Relationships:

1. $\bar{x}$ solves VIP $\Leftrightarrow \Psi_\alpha(\bar{x}) = 0$
2. $\bar{x}$ is fixed point $\Leftrightarrow \bar{x} - \frac{1}{\alpha}f(\bar{x})$ solves eq
3. Normal equation: $0 \in F(\bar{x}) + N_K(\bar{x})$